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Influence of system boundary conditions on the life cycle cost and carbon emissions of CO_2 transport

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ABSTRACT

One of the critical stages in carbon capture and storage (CCS) is the transport of CO_2 from the point of capture to the storage site. Besides pipeline transport, the present proposition of marine CO_2 transport is to make use of vessels that transport chemicals of similar physical properties to CO_2 such as liquefied propane gas (LPG) and liquefied natural gas. Through a life cycle analysis, this study examines the influence of system boundary conditions on the total cost and benefits of CO_2 transport using a reference LPG tanker. While the levelized cost of transport (i.e. shipping) can be intuitively obtained at 31.65–47.05$/t when LPG and CO_2 are assumed to be transported as cargos between two fixed ports, the total system-level CO_2 emission is contentious when the greater "interconnected" systems including LPG and a fictitious power plant life cycle are considered. Base on the analysis of the proposed most reasonable boundary condition, a break-even carbon price is further examined and found to be ranging from 39.37 to 22.32$/t-$CO_2$. While the boundary conditions can influence the break-even price of CO_2, such variations and the associated change in boundary conditions can be used to facilitate policy debate on the right carbon price for international shipping.

1. Introduction

The special report published by the Intergovernmental Panel on Climate Change (IPCC) suggests the possible devastating environmental impacts with a 1.5°C rise in the average annual global temperatures compared to pre-industrial levels due to the fast rising global carbon dioxide or CO_2 emissions (IPCC 2014). Amongst the various low carbon technologies, carbon capture, storage and utilisation (CCS/U) is believed to make an important contribution to the global decarbonisation effort (IEA 2018). Mikunda et al. (2021) used the Nilsson score assessment method to assess the interaction between CCS and the Sustainable Development Goals, and found that the deployment of CCS is a key driving factor in tackling climate change. Castrejón et al. (2018) used the Mexican energy model and found that when other energy efficiency measures are adopted, in order to achieve the national greenhouse gas emission goal, CCS technology is still needed. Although CCS technology can reduce the carbon dioxide emitted into the atmosphere, the carbon emissions produced by this technology itself cannot be ignored. When choosing low carbon technologies, emission reduction efficiency is a key consideration. Therefore, it is necessary to study the carbon emissions of CCS throughout the life cycle analysis.

Various studies have been conducted on the life cycle carbon emissions of CCS value chain. Using the life cycle inventory analysis method, Singh et al. (2015) found that after the power plant applies CCS technology, the system's material and resource usage has increased, and carbon emissions have been reduced. Liu et al. (2020) evaluated the energy consumption and atmospheric emissions of CO_2-EOR technology through the life cycle assessment method, and found that the use of this technology can increase China's contribution to climate change. Sathre et al. (2012) discussed the impact of seven key issues (eg. energy penalty, functional units, scale-up challenges, non-climate environmental impacts, uncertainty management, policy-making needs, and market effects) on CCS life cycle assessment. Considering the CCS life cycle carbon emissions, Singh and Colosi (2021) used the life cycle assessment method to evaluate the application value of the carbon capture, utilisation and storage (CCUS) platform by defining a standardized ratio metric – carbon return on investment. Kim et al. (2019) proposed a process-based method to study the life cycle net CO_2 reduction of an

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Ammonia CCS system, and discussed the impact of the capture, transportation, and storage processes on the CCS life cycle emission reduction effect. Viebahn et al. (2007) compared the life cycle assessment results of fossil-fired power plants and wind and solar thermal power plants with the consideration of life cycle carbon emissions of CCS.

In addition to focusing on the CCS value chain, the existing life cycle assessment also conducts cost research on parts of the CCS value chain. Seo et al. (2015) compared four carbon dioxide liquefaction systems based on life cycle cost analysis, and selected the optimal carbon dioxide liquefaction system. The transportation stage in the CCS value chain has recently received more research and attention. Kalyanarengan Ravi et al. (2017) used the mixed integer linear programming method to propose an optimal pipeline transportation model with CO_2-hub, which solves the problem of minimizing the life cycle cost of the CCS value chain transportation stage. Kjärstad et al. (2016) compared the cost or benefit of pipeline transportation and ship transportation by changing the distance and volume of carbon dioxide transportation, and obtained the respective applicable conditions for pipeline and ship through the break-even point.

Kim et al. (2019) shown that in the CCS value chain, carbon emissions during transportation account for about 23% of the total emissions throughout the life cycle of CCS. Transportation is an important link in the CCS value chain, but the pipeline transportation of carbon dioxide is now researched more comprehensively. While there are discussions about transporting CO_2 using ships, such as an LPG tanker (Aspelund et al., 2006), very few studies have examined the costs and benefits of CO_2 transport by sea. Decarre et al. (2010) analysed the feasibility of ship transportation, and studied the optimal cost of pipeline transportation and ship transportation under different transportation volumes and transportation distances. When the source and sink of CO_2 are not connected by land, transport by sea would be necessary. Therefore, there is still a need to further dissect the economic and environmental advantages and/or disadvantages of CO_2 transport in the CCS value chain to ascertain if long-distance transport of CO_2 should be encouraged by policy.

Lai et al. (2011) proposed a conceptual framework for evaluating green shipping practices and found that the environmental impact of the shipping industry cannot be ignored. In order to assess the carbon emissions of CO_2 transport using ship, the emissions of shipping itself should be considered. It has been shown that the share of shipping emissions globally has increased from 2.76% in 2012 to 2.89% in 2018 (IMO 2020). Existing studies suggest a potential of reducing up to 75% of a ship's CO_2 emissions (IMO 2009) such as speed reduction (European Commission 2013), and optimal operational strategies (Dulebenets, 2018; Dai et al., 2018). Besides the CO_2 emissions due to the fuel consumption, a ship is also associated with a life cycle carbon emissions from manufacturing to end-of-life scrapping (Gilbert et al., 2017), of which the process includes shipbuilding (Xie et al., 2017), operation (Corbett et al., 2009), maintenance (Styhre et al., 2017), and ship scrapping and recycling (Du et al., 2017). Jain et al. (2016) quantified the material flow of a single ship by using the on-board documentation to obtain information at the end of the life of the ship, and put forward suggestions for improvement of ship design, making ship recycling more environmentally friendly. Choi et al. (2016) studied the important role of economic cost-benefit analysis methods and life cycle analysis methods in the end-of-life recycling management of ships. While CO_2 transport can facilitate reduction of CO_2 from combustion of fossil fuels on land, there are uncertainties associated with the cost and life cycle CO_2 emissions when the CCS value chain is enabled by international shipping.

As a contribution to the literature, this study employs a scenario-based approach to evaluate the life cycle cost and CO_2 emissions of transporting CO_2 via a reference LPG tanker. The concept of interconnected systems as proposed in V. Nian (2016) is employed to evaluate the influence of system boundary conditions on the life cycle carbon emission accounting. The objective is to identify a reasonable basis for evaluating CO_2 reduction potentials when shipping is used as means of CO_2 transport. The boundary conditions are further used to facilitate the discussion of carbon price through which CO_2 transport could become commercially viable.

The rest of paper is organised as follows. Section 2 discusses the methodology for computing the life cycle costs and CO_2 emissions. Section 3 presents the data and assumptions used in the case study. Section 4 provides the case study results and their implications. Section 5 concludes the paper with recommendation for future research.

2. Methodology

2.1. Life cycle carbon emissions

A ship is a complex energy system with fuel supply, engine, air-conditioning, lighting and other on-board energy-consuming components (Yuan et al., 2019). A ship is also a part of a greater life cycle energy system from building and construction, ship operation and maintenance, to end-of-life management (Nian and Yuan, 2017). The shipping system was conceived from a system producing energy (V. Nian, 2016; Wang et al., 2018) and products (V. Nian, 2016) to a system delivering services (Wang et al., 2019). As such, shipping is defined as a system for providing transport services for passenger and cargo.

Based on the general system boundary conditions, a life cycle shipping system has both energy and non-energy inputs, and outputs carbon emissions over the lifetime. It is assumed that the system boundary of the shipping system can be formulated from three dimensions, including the boundary between the system and its environments, the boundary between the main system and its sub-systems, and physical and temporal boundaries formatting the main system (Nian et al., 2014).

In this study, the life cycle shipping system is assumed to be focused on vessels transporting LPG. Coral Alameda, recorded in Clarksons Research Shipping Intelligence Network (2022), is an LPG ship with a minimum tank temperature of −104 °C and a maximum tank pressure of 7 bar. The maximum tank pressure of the LPG ship recorded in Clarksons Research Shipping Intelligence Network (2022) can even reach 20 bar. For the liquid carbon dioxide ship, with reference to Georg Hegerland and Pande (2005), conditions close to the triple point (i.e., −56.6°C, 5.2 bar) can reduce the shipping cost. It can be seen that the temperature and pressure of the LPG ship is sufficient for the transportation of liquid carbon dioxide. Therefore, it is technically feasible to build or transform the LPG ship to the LPG/CO_2 ship. Assuming existing ships, such as those used for transporting liquefied propane gas (LPG) can be used for CO_2 (−56.6°C, 5.2 bar (Panaitescu et al., 2018)) transport, transporting CO_2 could translate to an added revenue stream of a shipping company. Noted that two different cargos (i.e. LPG and CO_2) are transported in the same ship tank, tank cleaning is usually required before loading the other type of cargo. This will increase the total operational cost. However, tank cleaning cost is relatively small compared to the transportation cost, which is not considered in this paper.

Adapting from earlier developments, the conceptual framework of a life cycle LPG shipping system would include shipbuilding, transporting LPG from export to import locations, and ship decommissioning at the end of life (Fig. 1). The carbon emissions are due to the use of energy and non-energy inputs as well as fuel combusted inside the ship's engines. In earlier developments, a "product" is defined and transformed across the process chain of the life cycle system so as to determine the life cycle system and boundary formations. In this study, the "product" is defined as shipping service which is the delivery of LPG or captured CO_2.

For the considered shipping system, the goal of its LCA study is different from the LCA study on LPG/CO_2 delivered. The life cycle carbon emissions of the delivered "product" (i.e. LPG and/or CO_2) should account for the emissions due to its production process. However, the life cycle carbon emissions of the considered shipping system do not take into account the emissions from the production and consumption of the

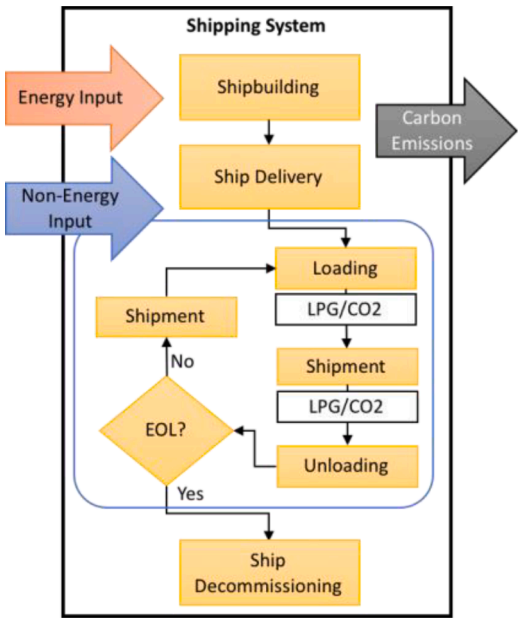


Fig. 1. Simplified conceptual schematic of life cycle LPG/CO2 shipping system. EOL refers to end-of-life.

delivered "product".

The proposition of using LPG ships to transport both LPG and captured CO_2 with a focus on the shipping system would lead to interconnections with the energy producing systems. A typical LCA study on LPG or CO_2 emissions due to (for example) electricity generation would involve an LCA main system for LPG or electricity production with respective system input-output and boundary conditions. These systems usually involve the transport of materials and/or fuels to support the activities within each process/stage of the life cycle system. An LCA study focusing only on the services provided by shipping would require additional set of boundary conditions.

The most relevant stage of the LPG life cycle system is transport from LPG production facilities to end users which usually require long-distance sea freight. The most relevant stage of a life cycle system for electricity production with carbon capture, storage and/or utilisation is the transport of captured CO_2 from the power plant to the geological storage or utilisation site. Building on the conceptual schematic as shown in Fig. 1 and with reference to the concepts as explained in V. Nian (2016), the interconnection amongst LPG production, electricity production and shipping systems can be formed by using LPG and CO_2 as the product across the system boundaries (Fig. 2). Since traditional electricity production generates a large amount of carbon dioxide by burning coal, it is selected as the capture source of carbon dioxide in this paper. The objective of such a formation of interconnected systems is to evaluate the total benefit and cost of CO_2 delivery by an LPG ship over the lifetime of the ship.

The objective as explained in the preceding paragraph leads to the formation of the "partial temporary boundary" around the shipping system which is synchronised based on the lifetime of the LPG ship. As such, the amount of LPG and CO_2 transported over the lifetime of the LPG ship is solely determined by the carrying capacity of the ship which is independent of the amount of LPG or CO_2 produced over the lifetime of their respective sources. The resulting physical boundary also allows for the evaluation of the total benefits in terms of revenue and decarbonisation.

A consistent argument of CO_2 transport for facilitating decarbonisation is the increase in carbon emissions when transporting LPG. Although LPG ships are not directly responsible for the CO_2 emissions due to the consumption of LPG, LPG ships are a vital part of the LPG supply chain and are deriving revenue from the transport services

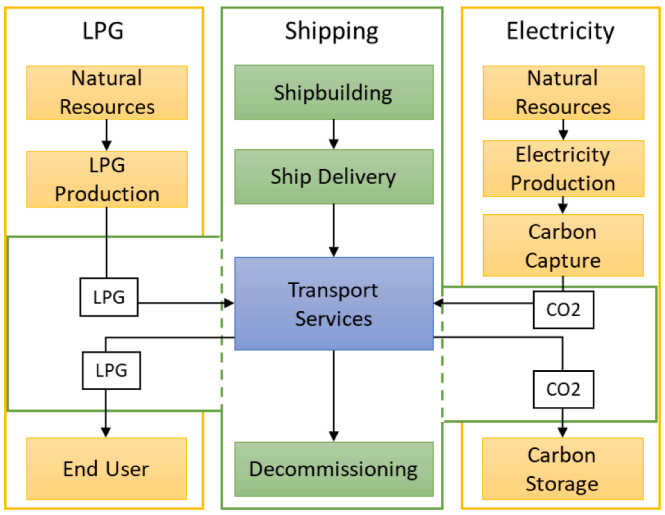


Fig. 2. Conceptual framework of system interconnections.

rendered. For consistency with the physical system boundaries of the interconnected systems, the net-CO_2 benefit of LPG-CO_2 transport is estimated in two approaches. One approach considers LPG and CO_2 only as cargo such that the life cycle carbon emission of the interconnected system is entirely due to the energy and non-energy input of the shipping system. The other approach considers transported LPG as a source for adding CO_2 indirectly to the interconnected system and transported CO_2 as a "sink" for subtracting CO_2 indirectly from the interconnected system.

The carbon emissions due to the fuel consumption take a large proportion of the life cycle carbon emissions of the shipping system. The fuel consumption can be computed based on the work done by the engine and the fuel consumption rate of the engine. The work done by the engine, W, can be expressed as

$$W = P \cdot T \cdot \sigma \tag{1}$$

where P denotes the engine power. T denotes the lifetime of ship. σ denotes the ship's utilization rate.

The ship's fuel consumption can be assessed based on the marine engine power output. The main engine power is usually determined through specific maximum continuous rating (SMCR) (Man Diesel and Turbo 2013). Four ships shown in Table 1 are taken as examples which could be used to transport LPG. Based on the above conceptual assumptions, they can transport both LPG and carbon dioxide in this paper. It can be found that different ship sizes as measured by cubic metre have different design speed, main engine power and auxiliary engine power.

With reference to Corbett et al. (2009), The instantaneous main engine power P_i at a given speed ν_i for ith trip can be expressed as

$$P_i = P_M \cdot \left(\frac{\nu_i}{\nu_0}\right)^3 \tag{2}$$

where P_M is the rated main engine power. ν_0 denotes the design speed.

Table 1
Design speed and engine power of ship.

Ship size (cubic metre)	Maximum speed (knots)	Main engine power (kW)	Auxiliary engine power (kW)
27,566	16	11,700	2112
38,405	15.5	10,680	3120
60,000	16.5	11,292	4800
87,749	16	12,800	5520

Data sources: https://sin.clarksons.net.

The engine work done per trip for the main engine and auxiliary engine can be expressed respectively as

$$W_{M,i} = P_i \cdot \varphi_M \cdot \frac{d}{v_i} \tag{3}$$

$$W_{A,i} = P_A \cdot \varphi_A \cdot \frac{d}{v_i} \tag{4}$$

where $W_{M,i}$ and $W_{A,i}$ represent the work done by the main and auxiliary engines respectively in a single voyage. P_A denotes the auxiliary engine power; φ_M and φ_A represent the loading factor of main engine and auxiliary engine respectively. d represents the distance between the exporting and importing ports.

The fuel consumption F_i for ith trip can be expressed as

$$F_i = W_{M,i} \cdot r_M + W_{A,i} \cdot r_A \tag{5}$$

where r_M and r_A denote the average fuel consumption rate of the main engine and the auxiliary engine respectively. According to Wang et al. (2007), the hourly fuel consumption rates for the main engine and the auxiliary energy are 206 g/kWh and 221 g/kWh respectively. The average load factor for the main engine is 0.8 and for the auxiliary engine is 0.5. Here the intermediate fuel oil (IFO) is taken as the bunker fuel. The carbon emission factor of IFO is 3.1 t-CO_2/t-IFO.

Besides the energy consumptions of the marine engines, the shipping system also includes the energy and non-energy inputs. The energy and non-energy inputs for each process (e.g. shipbuilding) can be expressed as

$$E_n = \sum_{j=1,2,\cdots} E_{n,j} \tag{6}$$

$$NE_n = \sum_{k=1,2,\cdots} NE_{n,k} \tag{7}$$

where E_n and NE_n denote the energy and non-energy inputs to the system for nth process. $E_{n,j}$ denotes the energy input by type j (e.g. diesel, electricity) for nth process. $NE_{n,k}$ denotes the non-energy input by type k (e.g. chemicals, metals) for nth process.

For the complex shipping system, there are usually a number of different processes. The total system energy input E_{sys} and non-energy input NE_{sys} is the sum over all processes which can be expressed as

$$E_{sys} = \sum_{n=1,2,\cdots} E_n = \sum_{n=1,2,\cdots} \left(\sum_{j=1,2,\cdots} E_{n,j} \right) = \sum_{n=1,2,\cdots} \left(p_n \cdot \sum_{j=1,2,\cdots} e_j \right) \tag{8}$$

$$NE_{sys} = \sum_{n=1,2,\cdots} NE_n = \sum_{n=1,2,\cdots} \left(\sum_{k=1,2,\cdots} NE_{n,k} \right) = \sum_{n=1,2,\cdots} \left(p_n \cdot \sum_{=1,2,\cdots} ne_k \right) \tag{9}$$

where p_n denotes the "product" made by nth process. For the proposed shipping system, the "product" refers to the amount of LPG/CO_2 transported over the ship's lifetime. e_i and ne_i denote the energy and non-energy inputs per unit of p_n produced respectively.

Given the energy and non-energy inputs to the system, the process carbon emissions can be computed by

$$C_E = \sum_{n=1,2,\cdots} \left(p_n \cdot \sum_{j=1,2,\cdots} c_{e,j} \cdot e_j \right) \tag{10}$$

$$C_{NE} = \sum_{i=1,2,\cdots} c_{ne,i} \cdot NE_i = \sum_{n=1,2,\cdots} \left(p_n \cdot \sum_{k=1,2,\cdots} c_{ne,k} \cdot ne_k \right) \tag{11}$$

where C_E and C_{NE} denote the carbon emissions due to energy and non-energy inputs respectively. $c_{e,j}$ denotes the carbon content for jth type of energy input. $c_{ne,k}$ denotes the carbon content for kth type of non-energy input.

Based on the system boundary conditions discussed in Nian et al. (2014), only the carbon emissions due to the direct conversion of the system's inputs are included for the LCA of the shipping system, such as the carbon emissions due to the fuel consumption and other chemical/physical reactions. Therefore, in order to be consistent with the system boundary conditions, the carbon emissions due to the production of the system inputs are not taken into account. As there is no direct conversion of system non-energy inputs for the LCA of the shipping system, the carbon emissions due the non-energy inputs C_{NE} can be excluded and the life cycle carbon emissions of the shipping system can be computed as

$$\begin{aligned} C_{sys} &= C_E + C_{NE} + C_{Fuel} \\ &= \sum_{n=1,2,\cdots} \left(p_n \cdot \left(\sum_{j=1,2,\cdots} c_{e,j} \cdot e_j + \sum_{k=1,2,\cdots} c_{ne,k} \cdot ne_k \right) \right) + C_{Fuel} \\ &= \sum_{n=1,2,\cdots} \left(p_n \cdot \sum_{j=1,2,\cdots} c_{e,j} \cdot e_j \right) + C_{Fuel} \end{aligned} \tag{12}$$

where C_{sys} denotes the life cycle carbon emissions of the shipping system. C_{Fuel} denotes the life cycle carbon emissions due to the fuel consumption of shipping which can be computed by multiplying the fuel consumption (see Eq. (5)) and the carbon emission factor.

2.2. Carbon accounting approach

As discussed in Section 2.1, different system boundary conditions can be considered. One approach considers LPG and CO_2 as cargo and that the life cycle carbon emissions of the interconnected systems are entirely due to the life cycle energy and non-energy inputs to the shipping system. The life cycle carbon emission factor of the main system can be expressed as

$$EF_{sys,1} = \frac{C_{sys}}{Q} \tag{13}$$

where Q represents the quantity of the cargo shipped.

The other approach as proposed considers LPG as both cargo and a source of CO_2 emission whose amount is computed using the carbon content of LPG at 82.5% by molecular weight percentage. Thus, every unit weight of LPG consumption can translate to 3 unit weight of CO_2 emissions after adding the weight of oxygen (O_2) to carbon (C) to form CO_2. The transported CO_2 is considered as both cargo and a "sink" of CO_2 emission where the transported weight is directly used for calculation. The life cycle carbon emission factor of the interconnected system in this approach can be expressed as

$$EF_{sys,2} = \frac{C_{sys} + 3 \times Q_{LPG} - Q_{CO2}}{Q} \tag{14}$$

where Q_{LPG} represents the quantity of LPG shipped and Q_{CO2} represents the quantity of CO_2 shipped. $Q = Q_{LPG} + Q_{CO2}$ in this case.

2.3. Net present value

A cost-benefit analysis (CBA) approach is proposed to evaluate both the life cycle cost and income of CO_2 and/or LPG transport. CBA is also known as the net present value (NPV) approach, which discounts future benefits and costs to current values (Arrow et al., 2013). The CBA approach has been used in conjunction with LCA studies as seen in (Nian et al., 2019; Nian and Zhong, 2020). It can easily quantify the results under different scenarios so as to better express the implications of the decisions for decision-makers (Arrow et al., 1996).

In the life cycle system discussed in this paper, the life cycle cost includes shipbuilding costs, operating costs, fuel costs, and decommissioning costs. Operating costs include port docking costs, crew member's salaries, spare parts costs and maintenance costs during

maintenance. Since there is no valid operating data for LPG and CO_2 carriers, in order to clarify the problem, the crude oil carriers will be selected as references for reasonable assumptions. The decommissioning cost of the large scale projects such as power plants as seen in (IEA 2017; IEA 2015) is used as reference for the shipping system. Therefore, the decommissioning cost is assumed to be 15% of the capital expenditure (CAPEX). The total discounted cost (TDC) over the lifetime of the ship can be expressed as

$$TDC = \sum_{t=0,1,2,\cdots} (INV_t + O\&M_t + FUEL_t + DECOMM_t - RESID_t)\cdot(1+r)^{-t} \tag{15}$$

where TDC represents the total discounted life cycle cost; INV_t represents the upfront investment costs for ship building; $O\&M_t$ represents the operating and maintenance costs (except fuel); $FUEL_t$ represents the fuel costs during shipping; $DECOMM_t$ represents the cost of decommissioning or scraping of the ship; $RESID_t$ represents the scrap or salvage value recoverable at the end of life, all at year t. r represents the discount rate.

The NPV of levelized cost of transport (LCOT) services as delivered by the LPG ship over its lifetime can be expressed as

$$LCOT = \frac{TDC}{\sum_{t=1,2,\cdots} Q_t\cdot(1+r)^{-t}} \tag{16}$$

in which

$$Q_t = Q_{LPG,t} + Q_{CO2,t} \tag{17}$$

where Q_t, $Q_{LPG,t}$, and $Q_{CO2,t}$ represent the quantity of all cargo, LPG and CO_2 respectively delivered at year t.

Freight charges for the provision of transportation services are the main revenue during the life cycle. Since there is no information on the price of CO_2 transport, the price of CO_2 transport is assumed to be correlated to that of LPG transport. The total discounted income (TDI) can be presented as

$$TDI = \sum_{t=0,1,2,\cdots} P_t\cdot Q_t\cdot(1+r)^{-t} \tag{18}$$

where P_t represents the unit price offered by shipping company for transporting LPG or CO_2.

The net profit and loss (PNL) of LPG and/or CO_2 transport in this approach can be expressed as

$$PNL = \frac{TDI - TDC}{\sum_{t=1,2,\cdots} Q_t\cdot(1+r)^{-t}} \tag{19}$$

3. Case study data and assumptions

According to the 2020 shipbuilding statistics, the 3 major shipbuilding countries are shown in Table 2. In the shipbuilding stage, only energy inputs such as fossil fuels and electricity are considered. All non-electrical energy consumption is reported as tonnes of coal equivalent (tce). During the construction of the ship, there are mainly six departments consuming energy, including production, coating, outfitting, assembly, production support and logistics whose energy consumptions based on per unit of lightweight tonnage (LWT) are reported in Table 3 (Nian and Yuan, 2017). The unit of non-electric energy consumptions is transferred to tonne of coal equivalent (tce).

Table 3
Shipbuilding energy consumptions for different departments.

Department	Electricity (kWh/LWT)	Other energy (tce/LWT)
Production	104.60	0.03
Coating	1.53	< 0.001
Outfitting	0.04	0.06
Assembly	96.36	0.004
Production Support	0.96	–
Total Energy Consumption	203.49	0.10

Upon construction, the ship needs to be transported from the country of construction to the country of exporting goods which include either the LPG or CO_2 exporter in this study. An immediate CO_2 transport need is the use of CO_2 for enhanced oil recovery (EOR) (Lake and Bryant, October 2018). It is possible for LPG and CO_2 exporters to form a complete close-loop economy as LPGs are usually produced from oil exporting countries which would need EOR at some stage. Once the LPG cargo is delivered to LPG importer, the same port could reload the LPG tanker with liquid CO_2 and export it back to the LPG exporter for the EOR or geological storage. This could be the most economical arrangement for both LPG exporter and importer to maximise the usage of marine fuel. In this study, it is assumed that the same ship would repeat the same course of travel throughout the life cycle.

The life cycle of a ship also includes maintenance downtime, loading and unloading time, manoeuvre time and waiting time, in addition to sailing time at sea. The average maintenance downtime is 33.89 days annually and the loading and unloading time and waiting time for each voyage are 3 days and 1 day respectively (Chatzinikolaou and Ventikos, 2015). Manoeuvre time accounts for 4% of each voyage time. The times of loading and unloading for LPG and CO_2 are assumed to be the same. Therefore, the loading and unloading times for transporting both LPG and CO_2 are twice as many as transporting only LPG or CO_2. In addition, the transportation of liquefied gas needs to consider the effects of temperature and pressure. The density of liquefied petroleum gas and typical liquefied carbon dioxide are 0.493 t/m3 and 1.2 t/m3 respectively (Aspelund et al., 2006) when the temperature and pressure are suitable for ship transportation. As such, the total weight of CO_2 that can be loaded into the LPG tanker based on the assumed volumetric capacity exceeds the ship's deadweight. Thus, the maximum tonnage for CO_2 transport is capped at the LPG tanker's deadweight.

During the operation of the ship, the repair process of the ship is similar to the construction of the ship. According to Nian and Yuan (2017), the electricity and other energy consumed by the hourly maintenance are 0.067 kWh/t and 0.0003 tce/t respectively. At the end of the ship's lifetime, the ship will be scrapped and the useful materials will be recovered. Energy input is required in the material recovery process, and the energy consumption during the disassembly process is shown in Table 4.

The speed of the ship and the distance between ports has direct impacts on the life cycle carbon emissions, costs, and revenue of the ship. Based on Eq. (2), the fuel consumptions of vessels at different speed can be obtained as shown in Fig. 3. It can be found that vessels with different capacity values have the lowest energy consumption at a speed of around 8 knots. Since the tanker segment has been promoting larger sizes of vessels, the vessel with the largest capacity is chosen as the reference case study, and the optimal speed is set to be 8 knots.

Table 2
Major shipbuilding countries as of 2020.

Countries	Completed Gross Tonnage (million)
South Korea	49.60
China	43.90
Japan	13.05
Others	5.00

Table 4
Energy consumption of ship decommissioning for different fuel types. MJ refers to mega joule.

Fuel type	MJ/LWT
Liquefied propane gas	753.62
Diesel	130.78
Electricity	0.9

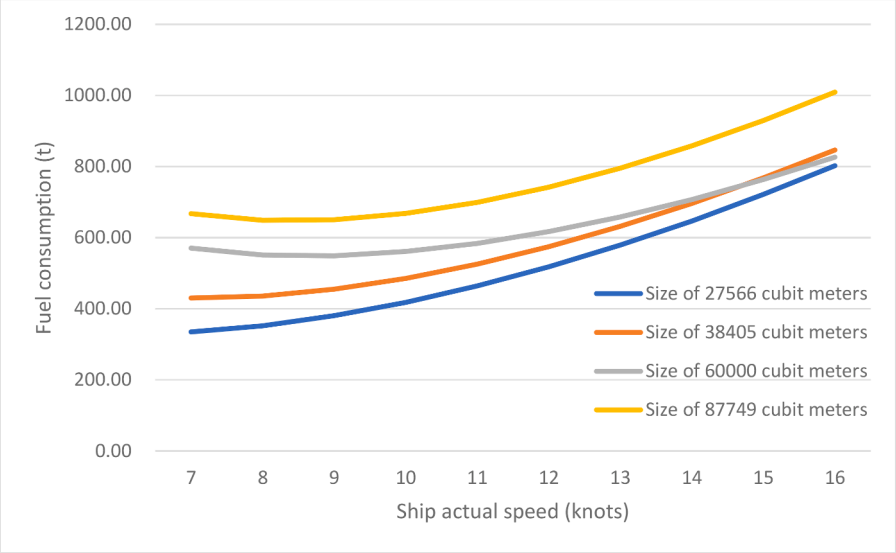


Fig. 3. Effect of ship speed on fuel consumption for different ship sizes.

In order to generalise the problems under investigation, Saudi Arabia, which produces a large amount of oil, is regarded as an exporter of LPG and an importer of carbon dioxide. China, which imports a large amount of oil and emits a large amount of carbon dioxide, is regarded as an importer of LPG and an exporter of carbon dioxide. China's total shipbuilding volume ranks amongst the top in the world, so China is regarded as a shipbuilding country in the case. The grid emission factors of Saudi Arabia and China are 709 g-CO_2/kWh and 623 g-CO_2/kWh respectively, and the distance between them is 5938 nautical miles.

The construction costs of the ship are shown in Table 5. Since no effective LPG/CO_2 carriers operating costs have been found, the operating costs of the crude oil tanker will be substituted here. The operating cost is $ 7116 per day. With reference toClarksons Research Shipping Intelligence Network (2022), the ship's scrap revenue is 428$/LWT, the average fuel price in 2019 is 348$/t, and the LPG carrier's freight rate is 67.25$/t.

4. Results

Combining the conceptual framework of Fig. 2, there are four possible ways in calculating the total CO_2 emissions of the interconnected system. The first way is to account for the CO_2 emissions only due to the shipping system. Both LPG and transported CO_2 are considered as cargo only. In this way, there are three possible scenarios including that only CO_2 is transported, only LPG is transported and both CO_2 & LPG are transported. In the second way, CO_2 emissions due to the shipping and the fictitious power plant system are accounted for in the total system-level CO_2 emissions. This is the equivalent of the temporal expansion of the physical boundary to include CO_2 as a "product" of the two systems interconnected at the shipping stage. Similarly, in the third way, CO_2 emissions due to the shipping and the LPG system are accounted for in the total system-level CO_2 emissions. The final way is to include the CO_2 emissions of all three interconnected systems, namely, shipping, LPG, and power generation systems.

Table 5
New building price of the ship.

Ship size (cubic metre)	New building price (M$)
27,566	57
38,405	42.5
60,000	81
87,749	120.59

4.1. CO_2 emissions from shipping system

4.1.1. Only transport CO_2

In this scenario, the ship transports liquid carbon dioxide from the export port (China) to the import port (Saudi Arabia). After the carbon dioxide is unloaded, the empty ship returns to the export port for the next transportation. The evaluation results for this transportation scenario are shown in Table 6. At a commercial discount rate of 7%, from the perspective of life cycle analysis, the total carbon emissions generated by ship navigation are 0.67 million tonnes (mt). The total volume of liquid carbon dioxide transported is 7.43mt. The total cost of transportation is 187.09M$. The total revenue from the provision of transportation services is 221.19M$. From the perspective of unit transportation volume, the unit cost of liquid carbon dioxide is 56.88$/t, and the unit profit is 10.37$/t. At a commercial discount rate of 15%, the total carbon emissions and the total volume of liquid carbon dioxide transported remain the same. The total cost of transportation is 157.80M $. The total revenue from the provision of transportation services is 125.79M$. The unit cost of liquid carbon dioxide is 84.36$/t, and the unit profit is −17.11$/t. The negative profit is due to the assumption that the ship returns to the export port with an empty tank at high discount rate. Shipbuilding costs that are not affected by the discount rate account for a high proportion of the cost, so the cost is less affected by the discount rate. The freight is affected by the depreciation rate, so there will be profits and losses (PNL) under two discount rates. When only CO_2 is transported, the life cycle carbon emission factor of the system is 0.09 t-CO_2/t-cargo, which means that every ton of CO_2 transported will produce 0.09 tons of carbon dioxide emissions. From the above evaluation results, it can be found that the transportation of carbon dioxide will cause losses to the shipping company under 15% discount rate. Therefore, it may not be profitable to develop the shipping

Table 6
Evaluation results for shipping system when only CO_2 is transported.

Item	Result	
	7%	15%
Total carbon emission (t)	668,812.66	668,812.66
Life cycle carbon emission factor (t-CO_2/t-cargo)	0.09	0.09
Total liquid CO_2 transported (t)	7,431,422.11	7,431,422.11
Total cost(M$)	187.09	157.80
Total revenue (M$)	221.19	125.79
LCOT ($/t)	56.88	84.36
PNL ($/t)	10.37	−17.11

business of liquid carbon dioxide for shipping companies.

4.1.2. Only transport LPG

In this scenario, the ship transports LPG from the export port (Saudi Arabia) to the import port (China). After the LPG is unloaded, the empty ship returns to the LPG export port for the next transportation. The evaluation results are shown in Table 7. The negative PNL under 15% discount rate is due to the assumption that the ship returns to the export port with an empty tank. The reason for the difference in profit and loss under the two discount rates is the same as Section 4.1.1. When only LPG is transported, the life cycle carbon emission factor of the system is 0.104 t-CO_2/t-cargo, which means that every ton of LPG transported will produce 0.104 tons of carbon dioxide emissions. From the above evaluation results, it can be found that if the ship only transports LPG and the return journey is empty, the LPG transportation business will cause the shipping company to lose money under 15% discount rate.

4.1.3. Transport both CO_2 and LPG

In this scenario, the ship transports LPG from the LPG export port (Saudi Arabia, import port of CO_2) to the LPG import port (China, export port of CO_2). After unloading the LPG, the ship will load the liquid carbon dioxide at the LPG import port, and transport it to the LPG export port to unload the carbon dioxide. After this, the ship will perform the next transportation. The evaluation results for this transportation scenario are shown in Table 8. It can be seen that the total carbon emission and the total cost for the dual-cargo situation (i.e. both LPG and CO_2 are transported) are slightly less than the one-cargo situation (i.e. only LPG or CO_2 is transported). This is because the number of trips for the dual-cargo situation is less due to the additional loading and unloading procedure. This will lead to less fuel consumption and cost for the dual-cargo situation. It can also be found that if the ship transports both liquid carbon dioxide and LPG, the transportation business can make the shipping company profitable. As shown in Fig. 4, the benefits of CO_2 only transported and LPG only transported do not have significant difference. When both CO_2 and LPG are transported, the benefits will rise significantly, and the life cycle carbon emission factor will decrease.

4.2. CO_2 emissions from interconnected systems

4.2.1. CO_2 emissions from shipping and power generation

Transporting carbon dioxide to storage places can reduce carbon emissions, but the life cycle carbon emissions from shipping should also be considered. The life cycle carbon emissions from shipping and power generation $C_{sys\&ele}$ can be expressed as:

$$C_{sys,ele} = C_{sys} - Q_{CO2} \tag{20}$$

where the life cycle carbon emissions of the shipping system C_{sys} can be computed according to Eq. (12). The quantity of the CO_2 transported during the life time of the ship Q_{CO2} can be computed by ship capacity multiplying the number of voyages.

As shown in Table 9, when the transported CO_2 is considered as cargo, the life cycle carbon emissions from shipping and power generation are 0.67mt. When the transported CO_2 is considered as "sink" for subtracting CO_2 indirectly from the system, the life cycle carbon emissions from shipping and power generation are −6.76mt which represents a reduction of 6.76mt of life cycle carbon emissions from shipping and power generation.

Table 7
Evaluation data for shipping system when only LPG is transported.

Item	Result	
	7%	15%
Total carbon emission (t)	668,812.66	668,812.66
Life cycle carbon emission factor (t-CO_2/t-cargo)	0.104	0.104
Total liquid LPG transported (t)	6,427,647.76	6,427,647.76
Total cost(M$)	187.09	157.80
Total revenue (M$)	191.31	108.80
LCOT ($/t)	65.76	97.54
PNL ($/t)	1.49	−30.29

Table 8
Evaluation results for shipping system when both CO_2 and LPG are transported.

Item	Result	
	7%	15%
Total carbon emission (t)	642,621.68	642,621.68
Life cycle carbon emission factor (t-CO_2/t-cargo)	0.048	0.048
Total cargo transported (t)	13,263,869.72	13,263,869.72
Total cost(M$)	185.81	157.07
Total revenue (M$)	394.79	224.51
LCOT ($/t)	31.65	47.05
PNL ($/t)	35.60	20.20

4.2.2. CO_2 emissions from shipping and LPG system

In the shipping and LPG system, the sources of carbon emissions include the life cycle carbon emissions of the shipping system and the carbon emissions generated by the combustion of LPG. The life cycle carbon emissions from shipping and LPG system can be expressed as:

$$C_{sys,LPG} = C_{sys} + 3 \cdot Q_{LPG} \tag{21}$$

where Q_{LPG} is the quantity of the LPG transported during the life time of the ship.

As shown in Table 9, when the transported LPG is considered as cargo, the life cycle carbon emissions from shipping and LPG system are 0.67mt. When the transported LPG is considered as "source" for adding CO_2 indirectly to the system, the life cycle carbon emissions from shipping and LPG system are 19.95mt.

4.2.3. CO_2 emissions from fully interconnected systems

The life cycle carbon emissions from the fully interconnected systems can be expressed as:

$$C_{sys,full} = C_{sys} + 3 \cdot Q_{LPG} - Q_{CO2} \tag{22}$$

As shown in Table 9, when the transported LPG and CO_2 are both considered as cargo, the life cycle carbon emissions from interconnected systems are 0.64mt. When the transported CO_2 is considered as "sink" for subtracting CO_2 indirectly from the interconnected system and the transported LPG is considered as "source" for adding CO_2 indirectly to the interconnected system, the life cycle carbon emissions from interconnected systems are 11.99mt.

4.3. The break-even carbon prices

In order to alleviate the greenhouse effect, carbon tax policy has been introduced. The carbon tax policy will affect the cost and income of enterprises, and further affect the profitability of the transportation system. The break-even carbon price refers to the carbon price that makes the life cycle net profit or loss of the transportation system zero. The break-even carbon prices are computed for the interconnected systems under different boundary conditions, which can be used as reference to make decision on carbon tax.

When only CO_2 is transported, the interconnected system would be formed by using CO_2 as the product across the system boundary. The total net profit or loss of the system can be expressed as:

$$TPNL_{CO2} = TDI - TDC - C_{Fuel} \cdot p + Q_{CO2} \cdot p \tag{23}$$

where *TPNL* represents the total net profit or loss of the system. p represents the carbon price. $C_{Fuel} \cdot p$ denotes the carbon cost due to the fuel consumption during shipping. $Q_{CO2} \cdot p$ denotes the benefit due to the transported CO_2.

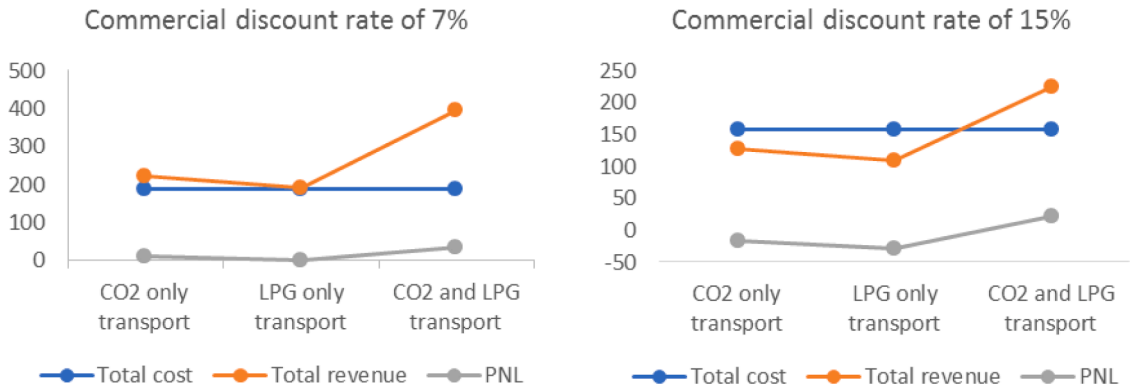


Fig. 4. Evaluation results for shipping system under three transportation scenarios.

Table 9
CO_2 emissions from interconnected systems under different boundary conditions.

System	Approach	CO_2 emissions (t)
Shipping and power generation	cargo	668,812.66
	cargo and "sink"	−6,762,609.44
Shipping and LPG system	cargo	668,812.66
	cargo and "source"	19,951,755.93
Interconnected system	cargo	642,621.68
	cargo, "sink" and "source"	11,985,160.13

The break-even carbon prices when CO_2 is transported are shown in Table 10. One setting is to consider the transported CO_2 as cargo. Eq. (23) changes to be $TPNL_{CO2} = TDI - TDC - C_{Fuel} \cdot p$. Under the discount rate of 7%, break-even carbon price is 113.97$/t. Under the discount rate of 15%, there is no break-even carbon price greater than zero to make the total net profit or loss of the system zero. In this setting, the loss of the shipping company will be greater if the carbon price is higher. The other setting is to consider the transported CO_2 as "sink" for subtracting CO_2 indirectly from the interconnected system. Under the discount rate of 7%, there is no break-even carbon price. When the carbon price is greater than zero, the total profit will be greater than zero no matter how the carbon price changes. Under the discount rate of 15%, the break-even carbon price is 18.86$/t. That is, the shipping company can become profit when the carbon price is greater than 18.86$/t.

When only LPG is transported, the LPG and shipping system would be formed by using LPG as the product across the system boundary. The total net profit or loss of the system can be expressed as:

$$TPNL_{LPG} = TDI - TDC - C_{Fuel} \cdot p - 3 \times Q_{LPG} \cdot p \quad (24)$$

The break-even carbon prices when LPG is transported are shown in Table 10. When the transported LPG is considered as cargo, the last term in Eq. (24), $3 \cdot Q_{LPG} \cdot p$, will not be taken into account. Under the discount rate of 7%, the break-even carbon price is 14.13$/t. Under the discount rate of 15%, there is no break-even carbon price, and the higher the carbon price, the greater the loss of the shipping company. When the transported LPG is considered as a source for adding CO_2 indirectly to the interconnected system, under the discount rate of 7%, the break-even carbon price is 0.48$/t. Under the discount rate of 15%, there is no break-even carbon price. When the carbon price is greater than zero, the total profit will be less than zero no matter how the carbon price changes.

Table 10
The break-even carbon price under different system boundary conditions.

Cargo transported	Settings	Break-even carbon price ($/t, 7%)	Break-even carbon price ($/t, 15%)
CO_2	cargo	113.97	–
	cargo and "sink"	–	18.86
LPG	cargo	14.13	–
	cargo and "source"	0.48	–
CO_2 and LPG	cargo	726.49	405.58
	cargo, "sink" and "source"	39.37	22.32

When both CO_2 and LPG are transported, the interconnection amongst LPG production, CO_2 production and shipping system would be formed by using LPG and CO_2 as the product across the system boundaries. The total net profit or loss of the system can be expressed as:

$$TPNL_{both} = TDI - TDC - C_{Fuel} \cdot p - 3 \cdot Q_{LPG} \cdot p + Q_{CO2} \cdot p \quad (25)$$

The break-even carbon prices when both CO_2 and LPG transported are shown in Table 10. When both transported CO_2 and LPG are treated as cargo, under the discount rate of 7%, the break-even carbon price is 726.49$/t. Under the discount rate of 15%, the break-even carbon price is 405.58$/t. When the transported LPG is taken as a source for adding CO_2 and the transported CO_2 as a "sink" for subtracting CO_2 indirectly from the interconnected system, under the discount rate of 7%, the break-even carbon price is 39.37$/t. Under the discount rate of 15%, the break-even carbon price is 22.32$/t.

From the results summarized in Table 10, it can be seen that when both LPG and CO_2 are treated as cargo, there are break-even carbon prices under 7% discount rate for different settings, while under 15% discount rate there is only a break-even carbon price for the setting that both LPG and CO_2 are transported. This is because the discount rate will affect the profitability of the shipping company. When the shipping company is profitable, it is reasonable to impose the carbon price on the shipping company for their emissions during transportation. Otherwise, the carbon price will further increase the shipping company's losses. As expected, the shipping company is able to afford higher carbon price when transporting both LPG and CO_2. When the transported CO_2 is considered as "sink" of the emissions, it can be seen that the higher carbon price is preferred when the system boundary only considers the transported CO_2. This is because the shipping company will be more profitable to transport CO_2 with the higher carbon price. The carbon price with 18.86$/t will make the shipping company from loss to profit under the 15% discount rate. Therefore, the carbon price can enhance the willing of the shipping company to transport CO_2. When the system boundary only considers the transported LPG, the carbon price will make the shipping company less profitable due to the "source" emission of LPG. When both LPG and CO_2 are considered in the system boundary, the total CO_2 emission is positive as the emissions of LPG is larger than the "sinked" CO_2 by transporting. Therefore, there are afforded break-even carbon prices for shipping company to transport both LPG and CO_2.

5. Conclusion

In this paper, the life cycle cost and CO_2 emissions of CO_2 transport

using a reference LPG tanker are computed via the concept of interconnected systems. Four system boundaries synchronisation is considered when computing the life cycle CO_2 emissions of the interconnected systems. The change in system boundary conditions can lead to quite some variations in the life cycle carbon emissions. Based on the proposed most rational boundary condition, the life cycle carbon emission factor of shipping is found to be 0.048t-CO_2/t-cargo when both CO_2 and LPG are considered as cargo. The costs of shipping are found to be 65.76 $/t-cargo under 7% discount rate and 97.54$/t-cargo under 15% discount rate when only LPG is considered as cargo and 31.65$/t-cargo under 7% discount rate and 47.05$/t-cargo under 15% discount rate when both LPG and CO_2 are considered as cargo. Arguably, adding CO_2 as additional cargo would improve the economics of shipping.

When the shipping sector is subject to a carbon price, findings from this study suggest 18.86$/t-$CO_2$ would be a desirable level of carbon price at which transport of CO_2 using international shipping can become commercially viable. Due to the limitation of the present methodology, it is difficult to contextualise the benefits of reduced carbon costs at the power generation side due to carbon capture and sequestration through CO_2 shipping. It is out of the scope of this paper to discuss the carbon price level at which power plant owners would be willing to consider carbon capture and sequestration. Based on the assumptions in this study, power generation can potentially save carbon costs based on the life cycle CO_2 amount transported to sequestration site noting the potential increase in capital and operating cost due to the deployment and operation of carbon capture technology. Noted that there are various uncertainties of the assumptions in the calculation such as the uncertainties of cost and fuel consumption, it would be of interest to verify the impact of these uncertainties. Therefore, another important future work is to use the proposed scenario based approach or other uncertainty quantification methods to assess the effects of different uncertainties.

Declaration of competing interest

The authors declare no competing interests.

Data availability

The data that has been used is confidential.

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